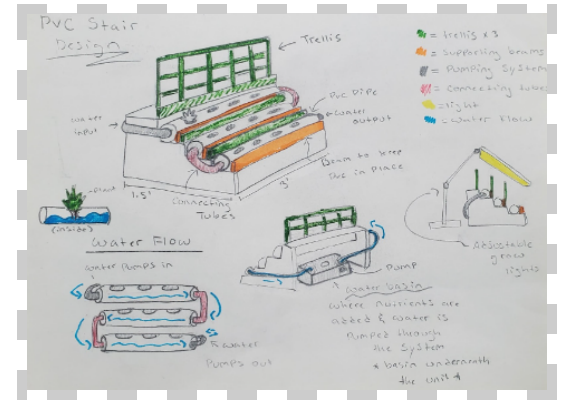


Element C

Presenting and Justifying of Solution Design Requirements

By understanding how the hydroponic and other hydroponics themselves work the team can develop a design that meets all the requirements given by our stakeholders. Some criteria needed for this project is that the control panel has to be connected to the pump and grow light, it has to be waterproof, and it has to be 30" by 60" in size. Our team has decided that it is vital to add some type of support for viny plants and taller plants so that plants can mature within the machine. As stated in this source hydroponic systems require supports to keep plants from falling over. The type of support needed depends on the type of plant being grown. Stakes, trellises, nets, and clips are the most common types of supports used in hydroponics. Wooden support is inexpensive and made from rot-resistant wood, while plastic support is lighter and easier to handle. Metal support is the strongest and ideal for heavy plants, but can break more easily. Bamboo support is expensive but can be made from natural materials. Clips are small devices that attach plant stems to a support structure and should be firmly attached to the stem. Other support materials are available for purchase depending on the look and weight of the plant ([SOURCE](#)). The team has consulted with our stakeholder who told us that the PVC pipe design would be the most effective in what she wanted our solution to be.



Design Features

- **Trellis:** Will be used to support viny plants such as peas and is going to be made out of plastic, and will be attached through a lego type of locking. They are also removable to be applicable to multiple plants.
- **Pump:** The submersible water pump will be used to transfer water from the reservoir to the top of the hydroponics machine, and will be used to add oxygen to the water which is vital to the plants.
- **Connecting Tubes:** Will be used to connect each layer and will be the main path the flow of water will follow.
- **Water Basin:** Is where the main source of water is located, the pump, and where the nutrient solution will be added in. Which will be located on the floor rather than the table.
- **Adjustable Grow Lights:** Grow lights are a specific type of electrical light that aids in plant growth. The purpose of the lights being adjustable is to create a better environment for different plant species or types to grow. To ensure they are meeting their individual needs the lights and temperature must be adjustable.
- **Beams:** The purpose of beams is to keep the PVC pipes in place and to support them to avoid breakage.
- **Nutrient Bottle and pH Holders:** Used as a storage component in the design to needed materials closer to the machine.
- **Baskets:** Used to hold the growing medium will be removable and interchangeable.
- **Plant Covers:** Will be used to germinate seeds by increasing humidity around seedlings.
- **Structure:** The structure or the "Staircase" aspect of the design to create an environment where the water can flow in a resourceful manner and reach all the plants to make sure they are properly emerged. It also allows for the space given to be used appropriately while also not restricting or minimizing the job of the machine.